



Build a Floating House



How can we build a house that floats safely in a flood?

Bangkok has lots of water, and floods are coming more often. Some families live in houses that float! Let's build one and test it.

TIME

60–75 minutes

GROUPS

Groups of 3–4

AGES

Starter 6–8 · Explorer 9–11

Learning goals

- ✓ I can build a raft that floats.
- ✓ I can test size, balance and materials fairly.
- ✓ I can use my results to design a better floating house.

STEAM in this activity

S

SCIENCE

Buoyancy: water pushes up on things that float.

T

TECHNOLOGY

Choosing materials that do not soak up water.

E

ENGINEERING

Build, test and improve a floating raft.

A

ART

Design a floating village.

M

MATHS

Count coins; Explorers find the middle of 3 tries.



Before the lesson

- ☐ Collect 6 plastic bottles with lids per group.
- ☐ Coins all the same size, about 30 per group.
- ☐ Tubs of water on the floor, towels, tape, card, foil trays.
- ☐ LEGO bricks for the house.



Safety

- Keep the water tub on the floor.
- Wipe up spills so nobody slips.
- Ask an adult if you need scissors.



Kit per group

- ☐  4 empty plastic bottles with lids
- ☐  Stiff card or a plastic tray for the floor
- ☐  Strong tape
- ☐  LEGO bricks for the house
- ☐  A big tub of water
- ☐  Towels for spills
- ☐  2 more bottles and tape for a small raft
- ☐  A bag of coins, all the same size
- ☐  The tub of water
- ☐  A foil tray about the same size
- ☐  A piece of cardboard the same size

Running order (70 minutes)

0–10 min Story: floods and floating homes.

10–30 min Build: bottle raft and LEGO house.

30–60 min Tests: size, balance and materials (10 minutes each).

60–70 min Share: design the best floating house.

The investigations

Start here: build: Build your floating house

Children record: Does your house float?; How many bottles did you use?; Draw your floating house *Explorers also:* How many LEGO bricks in your house?

What scientists often see: Floating houses are real! Some families in the Netherlands live in floating homes, and Thailand has raft houses on rivers. When a flood comes, the house rises with the water.

Test 1: Big raft or small raft?: Will a bigger raft carry more coins?

Children record: How many coins did it carry?; What happened at the end?; Draw it

What scientists often see: A bigger raft pushes more water out of the way. The water pushes back up, so a bigger raft can hold more. This push is called buoyancy.

Test 2: Middle or side?: Where should heavy things go so the house does not tip?

Children record: How many coins did it carry?; What happened at the end?; Draw it

What scientists often see: Weight on one side pushes that side down, so the house tips. Weight in the middle is shared by all the floats. Real floating houses keep heavy things low and in the middle.

Test 3: Best floating material: Which raft will carry the most coins?

Children record: How many coins did it carry?; What happened at the end?; Draw it *Explorers also:* After the test, is it soggy?

What scientists often see: Sealed bottles are full of air and do not soak up water. Cardboard soaks up water and goes soft. That is why real floating homes sit on sealed floats.

Talk about it

- Which raft carried the most coins? Why?
- Where should heavy things go in a floating house?
- Why did the cardboard raft fail?

What to look for

Starter: Counts coins carefully and says which raft carried more.

Explorer: Uses the middle of three tries and explains buoyancy or balance.

Art corner

Floating village. Draw a whole floating village: houses, gardens and bridges on the water.

Maths corner

Starter: Which raft carried more coins? Draw that many coins.

Explorer: Write your 3 tries in order: ____ ____ ____.
The middle one is ____.

Real-world link

Floating homes are real, in the Netherlands and on rivers in Thailand. They rise when floods come.

Going further

Add a roof garden to your house. How does the extra weight change the coins it can carry?

Curriculum links (for teacher review): NGSS 2-PS1-2 · NGSS 3-5-ETS1-3 (proposed)

WeLearn Growth Hub · Teacher guide

Activity 8: Build a Floating House